MANIPAL ACADEMY of HIGHER EDUCATION

Question Paper - Report

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Question Paper

MANIPAL ACADEMY OF HIGHER EDUCATION

B.Tech IIIrd Semester Mid term Examination September 2024

DIGITAL SYSTEM DESIGN [CSE 2123]

Marks: 30 **Duration: 90 mins.**

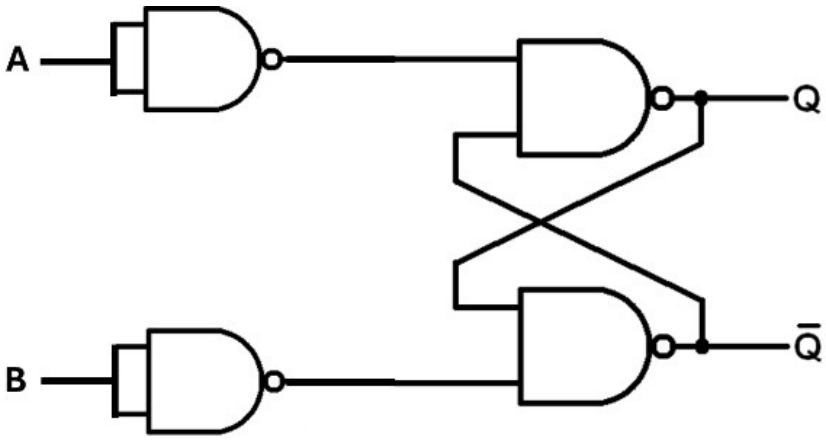
MCQ

Answer all the questions. Section Duration: 20 mins

1) A master slave flip-flop has the characteristics that: (0.5)

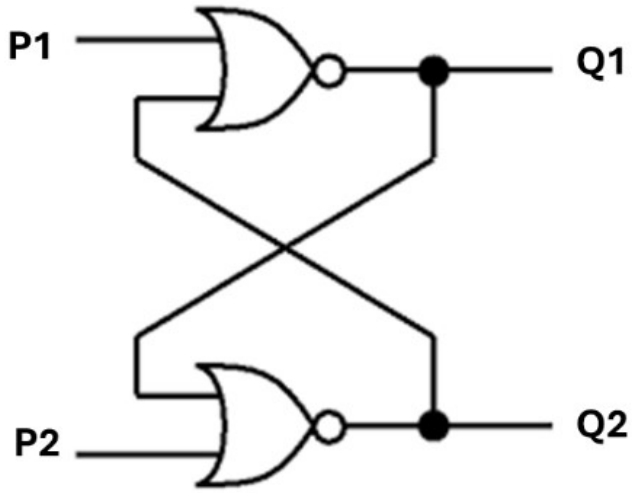
Change in the input immediately reflected in the output | Change in the output occurs when the state of the master is affected | Change in the output occurs when the state of the slave is affected | Both the master and the slave states are affected at the same time

2) The logic circuit represents (0.5)



S-R Latch. J-K Latch. S-R Flip-flop. J-K Flipflop.

3) Refer to logic circuit shown in figure. The inputs P_1P_2 for the logic circuit are first made 01 and then after a few seconds, made 11, the corresponding outputs Q_1Q_2 are (0.5)



First 01 then 00. First 11 then 10. First 10 then 00. First 01 then 01.

4) A system uses 16-bit unsigned integers. If X= 65520and Y=50, what would be the result of adding X and Y and how overflow should be handled? (0.5)

Result is 65570, no overflow detected. Result is 65570, overflow detected but ignored. Result is 14, overflow detected, and the carry-out is set. Result is 14, no overflow detected, and carry flag is set.

5) For an 8-bit carry-lookahead adder where the delay per gate is 4ns. If the sum requires the generation of the carry for each stage, what is the maximum delay for the carry to propagate through all 8 stages? (0.5)

16 ns 8 ns 12 ns 4 ns

6) The solutions to the quadratic equation $x^2-11x+22=0$ are x=3 and x=6. What is the base of the numbers (0.5)

10 9 8 6

7) A chemical plant needs a microprocessor-driven alarm system to warn of critical conditions in one of its chemical tanks. The tank has four HIGH (1)/ LOW (0) switches that monitor temperature (T), pressure (P), fluid level (L), and weight (W). Design a system that will notify the microprocessor to activate an alarm when any one of the following conditions arise: (0.5)

I. High fluid level with high temperature and high pressure

II. Low fluid level with high temperature and high weight

III. Low fluid level with low temperature and high pressure

IV. Low fluid level with high temperature and low weight

Then mention the output of the given system.

Y=TP+PL+TL $Y=TP+P\bar{L}+T\bar{L}$ $Y=T\bar{P}+P\bar{L}+T\bar{L}$ $Y=T\bar{P}+PL+T\bar{L}$

8) NAND is more sensitive to (0.5)

High input Low Input High Impedance None of these

9) Number of 2 X 1 multiplexers required to design 2^n X 1 multiplexer, are: (0.5)

2n 2^n 2n-1 2^n -1

10) Which of the following are building blocks of encoders? (0.5)

NOT gate AND gate OR gate NAND Gate

DESCRIPTIVE

Answer all the questions.

11) Write the characteristic equations for S-R, J-K, D and T FF. Design a flipflop whose characteristic table is shown below, using *J-K* flipflop: (4)

A	B	Q_{n+1}
0	0	1
0	1	Q_n
1	0	0

1	1	$\overline{Q_n}$

12) Design the logic for BCD addition adjustment. Draw the circuit diagram for a two-digit BCD adder using parallel adder. Write a Verilog code for one digit BCD adder using parallel adder structure. (3)

13) Design a two-bit comparator using logic gates and draw the logic diagram. Write a Verilog code for the same. (3)

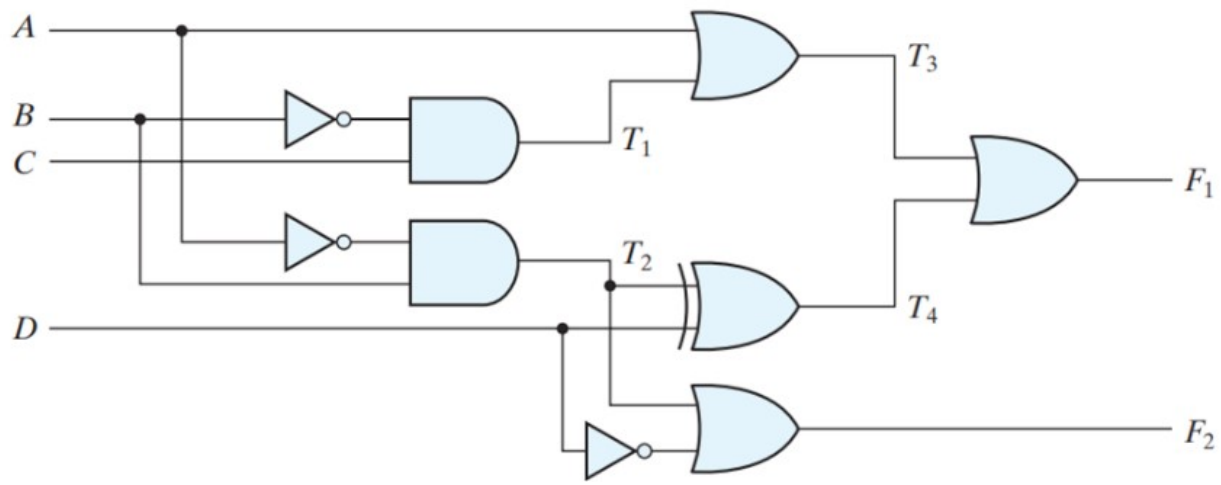
14) Design a 4-bit Gray-to-Binary code converter and implement the same using logic gates. (3)

15) For the function $f(w_1, w_2, w_3) = w_1w_2+w_1w_3+w_2w_3$, use Shannon's expansion to derive an implementation using a 2-to-1 multiplexer and any other necessary gates. (3)

16) A given system has four sensors that can produce an output of 0 or 1. The system operates properly when exactly one of the sensors has its output equal to 1. An alarm must be raised when two or more sensors have the output of 1. Design the simplest logic circuit that can be used to raise the alarm. (3)

17) Implement F (A, B, C) = Σm (1, 3, 5, 6) using (i) 4X1 multiplexer (ii) 2 x 1 multiplexer and a logic gate (2)

18) From the given gate level circuit in Figure .1. derive the Boolean expressions for T1 through T4. Evaluate the outputs F1 and F2 as a function of the four inputs. Find the simplified Boolean expressions for F1 and F2.



(2)

19) Derive the simplest Sum-of-products (SOP) and Product of Sum (POS) expressions for the function $f(x_1,x_2,x_3,x_4) = x_1\overline{x_3}\,\overline{x_4} + x_2\overline{x_3}\,x_4 + x_1\overline{x_2}\,\overline{x_3}$. (2)

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